



Tag Heuer x Super Mario

Tag Heuer will be releasing a \$2,150 Wear OS smartwatch in collaboration with Super Mario. <https://digit.in/aug21-15>



OnePlus caught throttling

OnePlus was caught throttling the performance of OnePlus 9 Pro "to save battery life". <https://digit.in/aug21-16>

menace by utilizing AI and video to prevent accidents, even before they occur. The DSM technology can determine if a driver is falling asleep at the wheel within 2-3 seconds and thus alerts the driver to pay attention. On the other hand, ADAS Forward Collision Warning Systems can alert the driver of an impending collision with a slower-moving or stationary car in front."

The cameras have been proven in the field in a most unique manner. The police in a satellite city of Mumbai are using the cameras to issue challans to traffic offenders, "the police vehicles in the Dombivli Kalyan Smart City have been fitted with our (Safe Cams') dash cameras, whereby the authorities can now issue challans to road offenders based on the camera footage. The large-scale use of dash cams and CCTV will ensure that traffic offenders are caught and taken to the court of law, which in turn will create a behavioural change over the long term."

The promise of autonomous cars is that they will reduce the number of road accidents. What we are seeing here is bits and pieces of the smartness and connectedness evolving rapidly, using the power of mobile computing and image processing, making sure that there is at least some intelligence in the vehicle whose sole focus is the road. Naidoo explains, "the greatest and popular safety devices like the seat belts, reverse cameras and ABS braking have all come through technological advancements. The next step is to add AI and sensors to safety devices so they can spot problems before they occur and take corrective action to prevent an accident. Video analysis means you have an additional 'digital eye' on the road that never sleeps or gets tired (unlike humans). We have introduced the DSM and ADAS cameras to India, and we are the first Indian brand to do so. This technology has been proven to reduce accidents and by around 2025 will

start to become standard on new vehicles in the EU."

The Driver State Monitoring (DSM) system used in the cameras is pretty sophisticated, and can provide a ton of useful information about the driver to fleet operators, even allowing them to establish a direct voice line over the device itself. Naidoo explains, "the DSM can identify within 2-3 seconds if a driver is falling asleep or is not paying attention to the road ahead and it rings an alarm to warn the driver. The Wi-Fi and 4G connectivity allow you to view any incident from anywhere in the world via our Cloud platform. In addition to this, the built-in GPS can help you as an owner to monitor your vehicle's driver and ensure they are staying within the required speed limits. Our AI uses the video data gathered by our cameras and runs an algorithm on it, so that face patterns can be detected accurately. The AI can identify a driver's eyes, ears and mouth, tracking them and other human gestures to identify if a driver is falling asleep or not paying attention to the road ahead."

Although they come for fleet and consumers, and in different feature options, there are some things that are common to all Safe Cams. They perform in the demanding conditions of India. Those who have used action cameras would be familiar with their tendency to heat up and stop recording. The components of the devices too have been selected to be suitable for local conditions. Naidoo says, "the 170 degree ultra-wide angle is the widest in the market (most Chinese brands are around 140 degrees) and the reason for this is in India, we have a lot of bikes that weave through traffic and the 140 degree lens will leave a blind spot on the side for the camera recording.

In terms of Low light sensors, we use high end Sony and OmniVision sensors – as most roads are poorly lit and there is not a lot of ambient light when driving at night. Our internal cabin cameras have IR lights so that you can see what is happening even in pitch darkness. All our equipment is tested up to 70 degrees Celsius in order to ensure that it will be suitable and most efficient even in the extreme hot climate. Whereas, for fleet equipment, we test up to 80 degrees Celsius conditions."



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MR VANESH NAIDOO
Founder & Director, Safe Cams

It's not just the hardware though, it's the pairing with intelligent software that makes this product something that addresses a major local issue, "with around 40% of accidents in India being attributed to drowsy driving, we feel our DSM devices can drastically reduce road accidents in India. Fleet owners will get a better understanding of their drivers and will be able to reward safer driving habits. With its low cost (will cost around 1% of a new commercial vehicle's value), companies can easily afford to make Indian roads safer as this cost can be recouped after preventing just one road accident or mishap."